

The second equality is enabled by the linearity of the scalar product, whereas the third equality follows from the orthogonality of the vectors $\mathbf{w}_k$: $\langle \mathbf{w}_k, \mathbf{w}_m \rangle = 0$ whenever $k \neq m$. Now, a formula for computing a_m has been reached:

$$a_m = \frac{\langle \mathbf{x}, \mathbf{w}_m \rangle}{\|\mathbf{w}_m\|^2}.$$

Ergo: For the representation of $\mathbf{x}$ as a linear combination of the basis vectors in an orthogonal basis, the coefficient in front of a given basis vector is the scalar product of that vector and $\mathbf{x}$, divided by the squared norm of the basis vector (Rydén, 2024).

Summary

To conclude: Any vector $\mathbf{x}$ in $\mathbb{R}^N$, where N is an odd number, can be represented in the following way (Rydén, 2024):

$$\mathbf{x} = a_0 \mathbf{f}_0 + \sum_{k=1}^M a_k \mathbf{f}_k + \sum_{k=1}^M b_k \mathbf{g}_k,$$

with

$$a_k = \frac{\langle \mathbf{x}, \mathbf{f}_k \rangle}{\|\mathbf{f}_k\|^2} \quad \text{and} \quad b_k = \frac{\langle \mathbf{x}, \mathbf{g}_k \rangle}{\|\mathbf{g}_k\|^2},$$

where

$$\|\mathbf{f}_0\|^2 = N \text{ and } \|\mathbf{f}_k\|^2 = \|\mathbf{g}_k\|^2 = N/2 \quad ; \quad k = 1, 2, 3, \dots, M.$$

The vectors may now be translated back to discrete functions, yielding the following equivalents of the recently summarized equations (Rydén, 2024):

$$x_n = a_0 \cos\left(2\pi \frac{0}{N}n\right) + \sum_{k=1}^M a_k \cos\left(2\pi \frac{k}{N}n\right) + \sum_{k=1}^M b_k \sin\left(2\pi \frac{k}{N}n\right) =$$

$$= a_0 + \sum_{k=1}^M a_k \cos \left(2\pi \frac{k}{N} n \right) + \sum_{k=1}^M b_k \sin \left(2\pi \frac{k}{N} n \right), \quad (23)$$

where

$$\begin{aligned} a_0 &= \frac{1}{N} \sum_{n=0}^{N-1} x_n f_{0,n} = \frac{1}{N} \sum_{n=0}^{N-1} x_n, \\ a_k &= \frac{1}{N/2} \sum_{n=0}^{N-1} x_n f_{k,n} = \frac{2}{N} \sum_{n=0}^{N-1} x_n \cos \left(2\pi \frac{k}{N} n \right) \quad ; \quad k = 1, 2, \dots, M, \quad \text{and} \\ b_k &= \frac{1}{N/2} \sum_{n=0}^{N-1} x_n g_{k,n} = \frac{2}{N} \sum_{n=0}^{N-1} x_n \sin \left(2\pi \frac{k}{N} n \right) \quad ; \quad k = 1, 2, \dots, M. \end{aligned}$$

Since the right-hand side of (23) is identical to that of (2), the ability of the latter to represent any real, discrete function with an odd number of points has been proven. It has been stated that this proof may be done, albeit with small modifications, also for even N . Doing so yields the following results (Rydén, 2024):

$$x_n = a_0 + \sum_{k=1}^{N/2} a_k \cos \left(2\pi \frac{k}{N} n \right) + \sum_{k=1}^{N/2-1} b_k \sin \left(2\pi \frac{k}{N} n \right), \quad (24)$$

where

$$\begin{aligned} a_0 &= \frac{1}{N} \sum_{n=0}^{N-1} x_n, \\ a_k &= \frac{2}{N} \sum_{n=0}^{N-1} x_n \cos \left(2\pi \frac{k}{N} n \right) \quad ; \quad k = 1, 2, \dots, N/2 - 1, \\ a_{N/2} &= \frac{1}{N} \sum_{n=0}^{N-1} x_n \cos \left(2\pi \frac{N/2}{N} n \right) = \frac{1}{N} \sum_{n=0}^{N-1} x_n \cos(\pi n), \quad \text{and} \\ b_k &= \frac{2}{N} \sum_{n=0}^{N-1} x_n \sin \left(2\pi \frac{k}{N} n \right) \quad ; \quad k = 1, 2, \dots, N/2 - 1. \end{aligned}$$

Comparing equation (24) to (3) yields the realization that they are indeed the same.